A balanced beam-splitter counterexample to the low-order Wigner–Rényi vacuum bound

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Abstract

The vacuum lower bound for Wigner–Rényi entropy holds for balanced beam-splitter states at orders $\alpha \geq 1/2$. We give a computer-assisted counterexample to its proposed extension to every $0 < \alpha < 1/2$. Both inputs are independently prepared in $(20|0\rangle + |2\rangle)/\sqrt{401}$, with mean photon number $2/401$ each. After balanced mixing and discarding one output, the retained Wigner function is exactly a Gaussian times the square of a quartic polynomial. At $\alpha = 1/4$, a directed full-plane integration bound gives $\int W^{1/4} < 9.43718$, whereas the vacuum integral is greater than 9.43892. The calculation treats polynomial zeros by an absolute-value chord and controls the entire Gaussian tail analytically. It uses exact physical reconstruction and two implementations of outward arithmetic. The result concerns one fixed order and one finite-support product input; it does not identify an optimum or a sharp boundary in Rényi order.

1 Question and contribution

For a state with a nonnegative Wigner function, differential entropy measures uncertainty over phase space. Van Herstraeten and Cerf proved a vacuum Wigner–Rényi entropy bound for the reduced outputs of balanced beam splitters with separable inputs at $\alpha \geq 1/2$ [1]. Their discussion proposes extending it to smaller positive orders. The physical restriction matters: general Wigner-positive states admit sub-vacuum entropy [2, 3], while the established beam-splitter Shannon bound remains valid. Purity-based sufficient conditions for the Shannon bound [4] address a different regime.

Here we test the proposed low-order extension inside the narrower class of independently prepared product inputs. The witness is an exact non-Gaussian ket of support $\{0, 2\}$ in each input. We reconstruct its reduced state and Wigner function algebraically, then give a full-plane upper enclosure at $\alpha = 1/4$ strictly below vacuum. No finite-box replacement of the entropy or perturbative approximation is used.

The counterexamples of Ref. [2] are explicitly outside the beam-splitter class; its two-level output-sector discussion concerns Shannon entropy. Our output has a nonzero four-photon component. Reference [3] also treats general Wigner-positive states and loss stability, rather than this fixed low-order, two-non-Gaussian-input witness. The contribution is therefore a restricted-channel counterexample, not another disproof of the general Shannon conjecture.

2 Conventions and exact physical state

We use dimensionless canonical quadratures, $[x, p] = i$, with $\hbar = 1$, $a = (x + ip)/\sqrt{2}$ and natural logarithms. Our phase-plane measure is $\mathrm{d}x\,\mathrm{d}p$, and $R = x^2 + p^2$. The vacuum Wigner function is

$$W_0(x, p) = \pi^{-1} e^{-R}. \tag{1}$$

For $\alpha > 0$, $\alpha \neq 1$, define

$$I_\alpha(W) = \int_{\mathbb{R}^2} W^\alpha \, \mathrm{d}x \, \mathrm{d}p, \qquad h_\alpha(W) = \frac{\ln I_\alpha(W)}{1-\alpha}, \tag{2}$$

with $0^\alpha = 0$. The vacuum satisfies

$$I_\alpha(W_0) = \frac{\pi^{1-\alpha}}{\alpha}, \qquad h_\alpha(W_0) = \ln \pi + \frac{\ln \alpha}{\alpha - 1}. \tag{3}$$

Prepare the two modes independently in

$$|\psi\rangle = \frac{20|0\rangle + |2\rangle}{\sqrt{401}}. \tag{4}$$

Its normalization is exact and $\langle a^\dagger a \rangle = 2/401$. The balanced lossless mixer acts on creation operators by $a^\dagger \mapsto (a^\dagger + b^\dagger)/\sqrt{2}$ and $b^\dagger \mapsto (a^\dagger - b^\dagger)/\sqrt{2}$. The convention differing by a sign of the second input gives the same state here because that input has even parity. The joint output is

$$|\Psi\rangle = \frac{1}{401}\left[400|00\rangle + 20|20\rangle + 20|02\rangle + \frac{\sqrt{6}}{4}(|40\rangle + |04\rangle) - \frac{1}{2}|22\rangle\right]. \tag{5}$$

Tracing out the second mode gives a density matrix on $\{0, 2, 4\}$,

$$\rho = \frac{1}{401^2}\begin{pmatrix} 160400 + 3/8 & 7990 & 100\sqrt{6} \\ 7990 & 1601/4 & 5\sqrt{6} \\ 100\sqrt{6} & 5\sqrt{6} & 3/8 \end{pmatrix}. \tag{6}$$

It is positive semidefinite as the Gram matrix obtained from Eq. (5), and its trace is one. The retained mean photon number is $2/401$. All higher Fock components vanish exactly: this support is part of the defined state, rather than a truncation of a Gaussian input.

The corresponding Wigner function is

$$W(x,p) = \frac{e^{-R}}{\pi} Q(x,p)^2, \qquad Q(x,p) = \frac{401 - 2R + R^2/2 + 20\sqrt{2}(x^2 - p^2)}{401}. \tag{7}$$

Consequently $W \geq 0$ everywhere, including its zeros, and $\int W \, \mathrm{d}x \, \mathrm{d}p = 1$. Positivity of the density operator follows from its preparation and partial trace, not merely from positivity of this phase-space expression.

Two algebraic constructions establish Eq. (7). The first transforms Eq. (6) using Fock-basis Weyl elements. The second starts with the Hermite wavefunction

$$\psi(q) = \frac{\pi^{-1/4} e^{-q^2/2}}{\sqrt{401}}\left(20 - \frac{\sqrt{2}}{2} + \sqrt{2}q^2\right). \tag{8}$$

Balanced rotation gives the joint wavefunction $\pi^{-1/2} e^{-(q_1^2+q_2^2)/2} P(q_1, q_2)/401$, where

$$P(q_1, q_2) = \left[20 - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}(q_1^2 + q_2^2)\right]^2 - 2q_1^2 q_2^2. \tag{9}$$

Partial tracing in q_2 and applying

$$W(x,p) = \frac{1}{\pi}\int_{\mathbb{R}} e^{-2ips}\langle x + s|\rho|x - s\rangle \, \mathrm{d}s \tag{10}$$

reduces to polynomial Gaussian moments. All coefficients can be compared exactly in $\mathbb{Q}(\sqrt{2})$. This also directly checks normalization. The coordinate calculation reproduces the known vacuum and identical one-photon input baselines; the latter gives $e^{-R}(R-1)^2/\pi$, including its nodal circle. The implementation details and exact outputs accompany the manuscript.

3 Full-plane integral bound at order one quarter

At the specified order,

$$I_{1/4}(W) = \pi^{-1/4} \int_{\mathbb{R}^2} e^{-R/4} \sqrt{|Q(x,p)|}\, \mathrm{d}x\, \mathrm{d}p, \qquad I_{1/4}(W_0) = 4\pi^{3/4}. \tag{11}$$

Fractional powers near polynomial zeros require explicit control. We use a cellwise upper inequality rather than estimating the error of point quadrature.

In polar coordinates $x = \sqrt{R}\cos\theta$, $p = \sqrt{R}\sin\theta$,

$$Q(R,\theta) = \frac{401 - 2R + R^2/2 + 20\sqrt{2}R\cos(2\theta)}{401}. \tag{12}$$

Each quadrant contributes equally and $\mathrm{d}x\,\mathrm{d}p = \frac{1}{2}\,\mathrm{d}R\,\mathrm{d}\theta$. For a cell $C = [a,b] \times [u,v]$ in the first quadrant, let

$$M_C = \int_C e^{-R/4}\,\mathrm{d}R\,\mathrm{d}\theta, \qquad J_C = \int_C e^{-R/4} Q(R,\theta)\,\mathrm{d}R\,\mathrm{d}\theta. \tag{13}$$

Suppose an outward enclosure gives $l \le Q \le h$ on C. An upper bound A_C on its weighted absolute-Q integral is

$$A_C = \begin{cases} J_C, & l \ge 0, \\ -J_C, & h \le 0, \\ \dfrac{h+l}{h-l} J_C - \dfrac{2hl}{h-l} M_C, & l < 0 < h. \end{cases} \tag{14}$$

The last case follows from the chord of the convex function $|q|$ on $[l,h]$. Its denominator is strictly positive. Cauchy–Schwarz with weight $e^{-R/4}$ then gives

$$\int_C e^{-R/4} \sqrt{|Q|}\,\mathrm{d}R\,\mathrm{d}\theta \le \sqrt{M_C A_C}. \tag{15}$$

Thus cells that straddle a zero are included without dividing by Q or assuming differentiability of its fractional power.

The cell moments are analytic. Set

$$m_0 = 4(e^{-a/4} - e^{-b/4}), \tag{16}$$

$$m_1 = 4[(a+4)e^{-a/4} - (b+4)e^{-b/4}], \tag{17}$$

$$m_2 = 4[(a^2+8a+32)e^{-a/4} - (b^2+8b+32)e^{-b/4}]. \tag{18}$$

Writing $\Delta\theta = v - u$ and $c_C = [\sin(2v) - \sin(2u)]/(2\Delta\theta)$, one obtains

$$M_C = \Delta\theta\, m_0, \qquad J_C = \frac{\Delta\theta}{401}[401 m_0 - 2m_1 + m_2/2 + 20\sqrt{2} c_C m_1]. \tag{19}$$

Natural interval evaluation of Eq. (12), with monotonicity of $\cos(2\theta)$ on $[0, \pi/2]$, encloses Q. The interior integral through $R = 100$ is at most $2\pi^{-1/4} \sum_C \sqrt{M_C A_C}$.

The omitted plane is controlled analytically. For $R \ge 100$,

$$|Q| \le 1 + \frac{2 + 20\sqrt{2}}{401} R + \frac{R^2}{802} < 2R^2. \tag{20}$$

The deliberately loose last inequality already follows for $R \ge 1$. It yields

$$I_{\mathrm{tail}} \le \pi^{3/4}\sqrt{2} \int_{100}^{\infty} R e^{-R/4}\,\mathrm{d}R = 16\pi^{3/4}\sqrt{2}\,26 e^{-25} < 1.929 \times 10^{-8}. \tag{21}$$

This bound includes all angles and all radii outside the disk.

3.1 Outward arithmetic and results

OpenAI Codex assisted with candidate discovery, algebraic derivation, certificate implementation and computational checking. Separate Codex sessions provided an additional coordinate-space reconstruction and an integer-interval implementation. The exact algebra and both certificates are supplied for reproduction; these AI-assisted checks do not replace human scientific review.

Uniform grids use 800×64 or 1600×128 cells in $[0, 100] \times [0, \pi/2]$. Radial endpoints are exact dyadic rationals. One implementation uses 40-digit directed Decimal arithmetic and a slightly tighter coefficient tail bound. A separately implemented 35-digit integer interval calculation uses Eq. (21) and reconstructs the state in coordinates without importing the first implementation. Both use the cell inequality above, so this comparison checks independent algebra and arithmetic implementations, not two unrelated integration theories.

In the integer implementation, additions and subtractions are exact; products and divisions are enclosed by integer floor and ceiling operations. Square roots use integer square roots with an upward correction when needed. The Machin identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ with alternating-series remainder encloses π. For e^x, $0 \le x \le 25$, terms through degree 160 are summed with a positive geometric bound on the remainder and the result is inverted to enclose e^{-x}. Sine and cosine use terms indexed 0 through 60 and an absolute next-term remainder on $[0, \pi]$. The remainders lie beyond the point where successive terms decrease. Every ensuing operation, including the sign-changing chord denominator, is outward rounded.

Arithmetic and grid	Full-plane upper bound	Tail upper bound
Decimal, 800×64	9.437261993640953	1.466×10^{-8}
Decimal, 1600×128	9.437179016856389	1.466×10^{-8}
Integer, 1600×128	9.437179021478335	1.929×10^{-8}

Table 1: Outward upper bounds on $I_{1/4}(W)$ including the entire tail. Decimal endpoints are from `alpha-quarter-jensen-enclosures.json`; integer endpoints are from `independent-enclosure.json`. The finer grid contains 621 sign-straddling cells. These values bound the integral from above; they are not two-sided enclosures of its value.

All three computations give a strict upper bound below the directed vacuum lower bound. In particular,

$$I_{1/4}(W) < 9.43718 < 9.43892 < 4\pi^{3/4} = I_{1/4}(W_0). \tag{22}$$

The last lower bound follows from the same enclosed π; its unrounded endpoint exceeds 9.4389219696587875. Since ln is increasing and $1 - \alpha = 3/4 > 0$, Eq. (22) proves $h_{1/4}(W) < h_{1/4}(W_0)$. One such normalized physical product input suffices to negate a statement quantified over every product input and every $0 < \alpha < 1/2$.

4 Physical meaning and limitations

Balanced mixing can produce a positive phase-space distribution from these independent non-Gaussian inputs while its quarter-order differential entropy lies below vacuum. Thus the channel's Wigner positivity and the established bounds at larger orders do not imply the proposed bound at all positive orders. The example has low mean occupation but contains an exact four-photon output component; discarding that component would define a different state. The measured quantity here is a Wigner–Rényi phase-space functional, not the von Neumann entropy of the reduced density operator.

This counterexample leaves the proved $\alpha \ge 1/2$ beam-splitter result, including its Shannon limit, intact. It does not establish a violating interval, an optimal deficit, a sharp order

threshold, a universal small-input expansion, or robustness under imperfect preparation and loss. The full-plane comparison is computer assisted: its trust base is the displayed inequalities, the supplied Python code and exact integer/rational or directed Decimal operations. No formal proof-assistant verification or experimental implementation is claimed. Comparison with primary versions available on 12 October 2026 found no matching restricted-class resolution; this is a bounded literature check and not an exhaustive priority assertion.

Data availability and reproducibility. The accompanying source package includes the exact reconstruction programs, both full-plane certificate implementations and their reference outputs. The accompanying instructions provide commands for exact Fock and coordinate reconstruction and both full-plane certificates in a scratch directory. All certificate programs require only Python's standard library. Exact rational interval endpoints and actual execution outputs are included. The table uses upper bounds only; no empirical search values are needed for the conclusion.

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